

Project Requirements — SmartPortal v2.0

Project Overview

SmartPortal v2.0 is a customer-facing web portal that enables enterprise clients to manage their accounts, view analytics, raise support tickets, and configure integrations. The project targets a Q2 2025 launch with a team of 8 engineers.

Functional Requirements

FR-01: Users must be able to register and log in using email or SSO (Google, Microsoft). FR-02: Dashboard must display real-time usage analytics with chart visualizations. FR-03: Support ticket system must support file attachments up to 25MB. FR-04: Integration marketplace must list and activate third-party connectors. FR-05: Admin panel must allow role-based access control with audit logging.

Non-Functional Requirements

NFR-01: System must handle 10,000 concurrent users with under 200ms response time. NFR-02: Uptime SLA of 99.9% with automated failover. NFR-03: All data must be encrypted at rest (AES-256) and in transit (TLS 1.3). NFR-04: GDPR compliance required — users can export and delete their data. NFR-05: Mobile-responsive design supporting iOS 15+ and Android 12+.

Technical Stack

Frontend: Next.js 14 with TypeScript and Tailwind CSS. Backend: FastAPI (Python 3.11) with PostgreSQL and Redis. Infrastructure: AWS ECS with RDS, ElastiCache, and CloudFront CDN. Monitoring: Datadog APM with PagerDuty alerting.

Timeline

Phase 1 (Weeks 1-4): Authentication, user management, and core dashboard. Phase 2 (Weeks 5-8): Support tickets, file uploads, and integrations. Phase 3 (Weeks 9-12): Admin panel, analytics, and performance optimization. Phase 4 (Weeks 13-14): QA, security audit, and production deployment.